

Lab 2 Report – Genetic Algorithm

Parameters

Distance threshold: 8000

Population size: 2000

Elitism rate: 0.1 (10%)

Tournament size: 8

Max generations: 10000

Representation

Each individual in a population is represented by an array of city indexes which tells the order in which the cities are being visited through the route.

Fitness function

Function which calculates total length of route from initial to final city. Evaluating this function tells the quality of fitness, with a higher length meaning worse fitness for this application. Every route length is calculated by adding all the precalculated distances from a matrix that are being taken in the specific route represented by the array. As well as adding the distance to the initial city.

$$Fitness = \sum_{i=1}^{n-1} \sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2}$$

Where n is the number of cities in an individual, x_i and y_i is the coordinates of a city, and x_{i+1} and y_{i+1} is the coordinates of the next city.

Selection method

To select individuals for reproduction, the Tournament Selection with a tournament size of 8 is used. Each randomly selected individual is then given different probability of being selected to be parent based on their fitness, with individuals with better fitness given a higher chance of being chosen.

Crossover

After the parents are selected, the model performs crossover with them to obtain new children. It uses crossover with order representation, which means that it directly copies a random section of parent A and fills the remaining route indexes with parent B in order. It will then switch the places of the same parents, parent A and parent B, to produce a second child.

Mutation

During mutations, two randomly selected cities in a child switch place to create a new mutated child. The rate of mutations changes dynamically during the process, at first the mutation rate is

high to maintain diversity and then slowly decreases by every generation to focus more on exploitation. The formula for the mutation rate is given by:

$$\text{Mutation rate} = 0.5 \cdot \left(1 - \left(\frac{\text{Generation}}{\text{Max generation}}\right)\right)$$

Replacement

To replace the old population, elitism is used to preserve the best individuals (10%) from the current population. These elite individuals are directly copied to the next generation. The rest of the new population is populated with the new children which were created from the crossover process. This ensures that the highest quality solutions are not lost during the evolutionary process.

Figures

